OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

MSc Biotechnology Thesis

Academic Year 2024-2025

Thesis title

Biosensors For Probiotic Bacterial Detection

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Acknowledgements

I would like to express my deepest gratitude to **Dr. Galina Brychkova** my supervisor, for their invaluable guidance, expertise, and encouragement throughout the course of this research. I am also grateful to the members of the Agriculture and Bioeconomy Research Centre for providing a supportive and collaborative environment that made this work possible.

Special thanks to the PhD students —**Aysel Malekpour and Pouya Motienoparvar**— for their generous help, insightful advice, and hands-on support during the experimental phases of my project. I sincerely appreciate their patience, technical expertise, and willingness to share their knowledge.

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Glossary of Abbreviations

CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats
ISR	Intergenic Spacer Region
MALDI	Matrix-Assisted Laser Desorption/Ionization
PAM	Protospacer Adjacent Motif
OD	Optical Density
CFU	Colony Forming Unit
PCR	Polymerase Chain Reaction
PBS	Phosphate-Buffered Saline
WGS	Whole Genome Sequencing
SAHARA	Split Activators for Highly Accessible RNA Analysis
SHERLOCK	Specific High-Sensitivity Enzymatic Reporter UnLOCKing
MRS	de Man, Rogosa and Sharpe medium

Abstract

Lactobacillus agilis is an understudied organism that exhibits potential for gastrointestinal colonization, metabolic adaptability, and pathogen suppression. However, its industrial use is largely restricted by absence of rapid, cost-effective, species-specific detection tools. This thesis reports a development of a biosensing platform that integrates pH sensitive natural dye based colorimetric system with CRISPR-Cas12a mediated molecular confirmation for *L. agilis* identification. The work started with optimizing culture conditions and creating a repeatable optical density (OD) – CFU curve to precisely standardize the viable cell input for biosensor calibration. In response to acidification during *L. agilis* growth, measurable spectrum shifts and visible color transitions are observed in bioink matrix prepared by using Anthocyanin as a biocompatible pH indicator. This metabolic signal, gave the first tier of detection. For the second tier, PCR primers targeting the species specific 16S-23S Intergenic Spacer Region (ISR) was experimentally validated and to improve sequencing accuracy, extended primers were created in silico for *L. agilis* LA3 reference genome to streamline the design of downstream CRISPR Guide RNA (crRNA). Although, tests to fully validate the Cas12a collateral cleavage assays remains a future step, this combined workflow demonstrated a proof-of-concept for a portable, affordable, and visually interpretable diagnostic tool. This platform can be used for quality control in the manufacture of probiotic foods and possibly for multiplexed detection of other spoilage microbes in food.

1. Introduction

Bacteria have been primarily linked with disease in humans, plants, and animals. However, due to advancements in the field of microbiology and immunology recently, this view changed dramatically, especially about the human gut microbes called probiotics (1). Increasing recognition for these "probiotic bacteria" is mainly because of the health benefits they offer, which includes gastrointestinal health, immune system modulation, and metabolic regulation. Consequently, a need to isolate and characterize new probiotic strains is intensifying due to consumer demand for natural food products driving the rapid expansion of the probiotic market globally.

1.1. Strain- vs Species-Level Detection in Probiotic Characterization

Critical findings from contemporary probiotic studies show that probiotic functionality is strain dependent; also, significant variations in metabolic profiles, host interaction, and health effects often vary across different strains within the same species, impacting both the efficacy and safety in host systems. For instance, *Lactobacillus rhamnosus* GG (ATCC 53103), a probiotic strain, exhibits a specific pilus structure that enhances the colonization potential by facilitating tight mucosal binding in the gut environment. These special pilus structures are absent or poorly expressed in other strains of *Lactobacillus rhamnosus* (2). Consequently, genotypic validation supporting strain-level taxonomic resolution is made as a prerequisite for the commercialization of probiotics and an important step in probiotic development (3).

Nevertheless, species-level identification remains an essential step when it comes to food environments for initial screening and quality control. In the context of industrial processes like probiotic validation and fermentation monitoring, rather than distinguishing between strains, confirmation on the presence or absence of a specific probiotic species is necessary (3). Especially while targeting probiotic species like *Lactobacillus agilis,* which are lesser known and may not have well-established strain libraries or strain-specific markers, species-level tools provide a cost-effective and more scalable approach (4). The demand for specialized staff, laboratories, and a large time commitment while using conventional detection techniques like PCR, Colony counting, and sequencing for probiotic species detection, highlights their drawbacks and the necessity for quick as well as inexpensive detection techniques (5).

1.2. *Lactobacillus agilis*, a Promising Yet Underutilized Probiotic Candidate

Weiss and associates recovered several homo-fermentative, meso-DAP-containing rod-like structures from municipal sewage in 1981 and identified that they could possibly belong to a distinct species within the *Lactobacillus* genus. In relation to the phenotypic and biochemical characteristics of the isolates, their work proposed two novel taxa, one of which is *Lactobacillus agilis*. The now catalogued DSM 20509^{T} (≙ JCM 1187, ATCC 43195) strain 262 is designated as the type strain (6). While the original strain came from the municipal sewage, subsequent studies showed the expanded ecological profile and adaptability of *Lactobacillus agilis* across a wide range of hosts and environments, which includes strains recovered from intestinal tracts of chickens, pigs, and various animals, and also from human feces, highlighting the ability of the species to thrive in diverse situations (7) (8) (9).

L. agilis exhibits a natural motility, a trait relatively rare among the Lactic Acid Bacteria (LAB). This motile behaviour of *L. agilis*, facilitated by the presence of functional flagella, serves a critical function within the Gastrointestinal System (GI). One of the key advantages of this *L. agilis* motile behaviour is its ability to perform chemotaxis–movement directed towards or away from specific chemical stimuli. *L. agilis* in the gut has been shown to exhibit positive chemotaxis towards mucin, which acts as a nutrient source and anchoring site for symbiotic and probiotic bacteria. Studies on this chemotactic response of *L. agilis* towards mucin significantly increases its ability to colonize and persist in the gut compared to other wild-type flagella-deficient mutant motile strains of *L. agilis* in vivo. These mutants displayed reduced interaction with the gut epithelium, resulting in overall diminished persistence in the gut environment (10). In probiotic applications, these traits are of significant interest as they may enhance gut retention and competitive exclusion of pathogens. Thereby *L. agilis* acts as a model for studying the motile behaviour of probiotics and to engineer colonization traits into other beneficial microbes.

Genomic insights into *L. agilis* reveal its metabolic versatility, especially in terms of carbohydrate utilization. For instance, *L. agilis* SNF7 genome harbors numerous genes encoding carbohydrate-active enzymes (CAZymes) like Glycoside Hydrolases (GHs) and Glycosyltransferases (GTs), reflecting the bacterium's adaptation to heterogenous nutrient sources, including dietary carbohydrates and host-derived glycans in the gut. This helps them to thrive in nutrient-variable gut niches and compete with pathogenic bacteria for energy (11).

Many *L. agilis* strains produce organic acids like lactic and acetic acid, lowering local pH, thereby inhibiting the growth of acid-sensitive pathogens like *E. coli* and *Salmonella*. In vitro studies of some *L. agilis* isolates, e.g., *L. agilis* KMS, show inhibition of E. coli growth through acid production and nutrient competition (Zhou et al., 2021). Certain strains of *L. agilis* are observed to downregulate NF-κB and MAPK pathway activation; these pathways, upon infection, upregulates pro-inflammatory cytokines like IL-6 and TNF-α as a gut immune response. By downregulating them, *L. agilis* reduces cytokine secretion and promotes anti-inflammatory cytokine production, leading to less tissue damage in in vivo models. Also, several *L. agilis* strains demonstrated increased tight junction protein expression, contributing to epithelial barrier strength and prevention of microbial translocation. Also, several strains of *L. agilis*, like SNF7 and 1003, showed a lack of hemolytic activity and susceptibility to a broad range of antibiotics, indicating the absence of an antibiotic resistance gene, which can be transferred to pathogenic bacteria in the gut, especially through mobile genetic materials like plasmids (11).

1.2.1. Lactobacillus agilis in Milk Derived Substrate:

In milk and fermented dairy products, *L. agilis* can contribute to microbial safety and can increase product safety through LAB's antimicrobial mechanisms. For instance, scientists discovered anti-*Listeria monocytogenes* bacteriocin from *L. agilis* C7 (agilicin C7). When screened using agar well diffusion assays, they exhibited string inhibitory activity against *Listeria monocytogenes.* Purified recombinant agilicin C7 (r. agilicin C7), obtained by cloning the gene for agilicin C7 into expression vector, also retained the anti listeria activity. r. agilicin C7 inhibited several *Listeria* species and was inactive against other gram positive and negative bacteria and fungi. It displayed a significant stability across wide range of pH, temperatures and in organic solvents, making it a suitable candidate for food application and also can be used in dairy where Listeria control is needed. Whole genome sequencing of *L. agilis* C7, also reveals three distinct bacteriocin gene clusters, Garvicin Q (activity against L. monocytogenes), Salivaricin A (activity against Micrococcus luteus), Blp-family Class II (competitive inhibition against closely related strains), suggests that *L. agilis* C7 have multi pronged anti microbial activity. This trait aligns with the trait of LAB, that says they improves quality and safety in dairy products by inhibiting pathogens during fermentation (12) (13) (14) .

According to taxonomy *L. agilis* is now a member of reclassified lactobacilli, with genomic characteristics compatible with probiotic potential and food bioprotective functions. The most logical use of *L. agilis* would be to use them as an adjunct in dairy products to speed up

acidification and to supress pathogen. *L agilis* is a promising addition to milk derived products, but its effects are strain specific and are needed to be tested on the intended dairy matrix for safety and performance (15).

1.2.2. Current Limitations in Detection and Validation

L. agilis is one of the lesser-studied members of the *Lactobacillus* genus, and it lacks extensive taxonomic infrastructure. This limits comparative genomic studies and the development of standardized detection assays for this species. Furthermore, *L. agilis* shows a high degree of similarity with other closely related *Lactobacillus* species. These phenotypic overlaps make it difficult to isolate *L. agilis* using traditional methods alone (16).

Especially in environments like the gastrointestinal tract, where multiple species of *Lactobacillus* coexist, studies on colonization and probiotic efficacy can be undermined if the species are misidentified. This highlights the importance of molecular specificity in taxonomic identification (17).

Although few molecular detection techniques are used to detect motile *Lactobacillus*, including *L. agilis*, there is a lack of assays that are species-specific, which limits the accurate detection of *L. agilis* in mixed microbial culture. This makes it challenging to study the dynamics of colonization in *L. agilis* and its probiotic efficacy *in vivo*.

1.3. Existing detection methods for lactic-acid bacteria – colony counts, qPCR, MALDI-TOF, metagenomics

Conventional microbiological, molecular, and sophisticated spectrometric methods are usually used for the detection of *Lactobacillus agilis*.

1.3.1. Culture-Based Colony Counting:

This method is the most standardized technique in microbiology, which involves culturing bacteria on agar plates under specific conditions (anaerobic or microaerophilic for *L. agilis*) and counting the colony-forming units (CFUs) manually or automatically using software. For selective growth of lactobacilli such as *L. agilis*, media like MRS agar is used generally. In addition to giving a direct measurement of viable bacteria count, it is also cost-effective and easy to implement. But this technique fails to detect bacteria that are present in very low numbers, in a viable but non-culturable (VBNC) state and morphologically identical to related species.

When researchers tried to replicate the natural rumen environment for further insights into the cow rumen-residing bacteria, they cultivated the bacteria under anaerobic conditions in a lab setting to create a rumen culture library with bacterial strains that could thrive in that setting. Even though *Lactobacillus* species were found and enumerated, they needed to determine the precise strains, which involves more complex techniques like genome sequencing. While examining all the DNA in a sample along with all the grown DNA (metagenomic data), they found that certain strains did not grow in the lab setting, implying that particular bacterial strains are not picked up by the culture, highlighting the drawback of conventional culturing. So, this makes this method not suitable for capturing the full diversity of the microbial population, especially those that require very special and strict conditions (18).

1.3.2. MALDI-TOF Mass Spectrometry (Matrix-Assisted Laser Desorption Ionization–Time of Flight):

MALDI-TOF is a proteomics-based method involving ionizing and vaporizing proteins in microbes, generally ribosomal, to create a distinct spectrum, which can be compared against managed databases to identify a species or genus.

This is one of the reliable techniques to get species-level identification if the target species is present in the reference database, and it is perfect for screening large quantities of microbial isolates from food and the environment. Also, a minimal amount of sample is sufficient.

This still requires colony isolation; hence it is subjected to the same restrictions as stated above (see section 1.3.1). Further, *L. agilis* can be misdiagnosed or reported as "no match" if it is not listed in the database; especially strains with substantially conserved protein profiles may be difficult to evaluate.

Studies focusing on feral chicken gut microbiota using MALDI-TOF for identifying bacterial isolates, show that MALDI-TOF is reliable when combined with other genomic tools. Since they are working with feral chicken gut microbiota, which is most likely to include undocumented strains and isolates that are not present in the MALDI-TOF database, they used 16S and whole genome sequencing to support the information obtained from MALDI, especially when deeper functional profiling is needed. While MALDI itself is rapid, it cannot resolve similar or novel species, so they used 16S for genetic confirmation and whole genome sequencing for functional gene annotation, creating a three-tiered validation pipeline to ensure reliable identification (19).

1.3.3. Metagenomics:

This involves DNA extraction and sequencing directly from biological or environmental samples. Rather than growing the microbes separately, this allows for the extraction, sequencing, and identification of all the DNA at once. This is very helpful in environments where microbes cannot be cultured using the standard lab techniques (20)

There are two types of metagenomics. One is amplicon-based, which focuses on a particular gene that is present in all the bacteria, e.g., the 16S RNA gene. Which involves DNA extraction from the sample, PCR amplification of the 16SRNA gene using universal primers, and sequencing of the amplified gene followed by comparison with a reference database to find the bacterial taxa. Even though it is efficient, it's limited to taxonomy only, and there will not be much information about the genes the microbes have. On the other hand, shotgun metagenomics, sequences not just one gene but all the DNA. The process involves extracting the total DNA and fragmenting it into small pieces and sequencing everything, followed by genome reconstruction and gene identification using computational tools. This technique can detect all organisms, including fungi and archaea, with high resolution but requires high-quality DNA, complex bioinformatics tools, and expensive equipment (21) (22).

1.4. Naturally Derived Indicators in Food Systems:

Now a days customer demand for clean label and alternatives to synthetic additives are growing, making the global food industry to turn towards naturally derived compounds and pigments. Plant pigments are highly valued for their food safety and environmental compatibility. Among these, anthocyanins – a flavonoid pigment, are studied vastly due to their ease of extraction, intense color range and pH responsiveness (23) (24).

1.4.1. Application of Anthocyanin in Smart Food Packing:

Incorporating anthocyanins into biodegradable polymer matrices has shown promising results for the development of smart food packing systems to monitor product quality in real time (25). They operate on the principal that protein rich foods release volatile nitrogen compounds like ammonia during microbial spoilage. The accumulation of these compounds elevates the local pH, triggering a visible color change in anthocyanins

Hashim et al. (2023) developed a novel pH – sensitive film to monitor freshness of chicken breast. The film was developed using agar methylcellulose matrix enhanced with anthocyanin

extract from purple Chinese cabbage. The film developed showed clear chromatic shifts in response to variations in pH, making it a promising natural indicator in smart food packing (26). In most cases the choice of polymer affects the stability and responsiveness of anthocyanins, Klaric et al. (2025) reviewed a range of biopolymers for compatibility with anthocyanin and concluded that biopolymers such as chitosan, starch, cellulose and gelatin are highly compatible and suitable for integration with anthocyanin (27). To diversify raw material supply, recent studies explored non traditional sources on anthocyanin, in which they developed cellulose chitosan composite films incorporated with anthocyanin derived from butterfly pea flowers and red cabbage. These films showed improved tensile strength, flexibility and pH response when compared against single source anthocyanin films and control films without anthocyanin. This indicates that both the pigment source and the matrix composition also play a major role in performance (28).

1.4.2. Advantages and Limitations of Anthocyanin

Anthocyanins are naturally occurring elements in food and vegetables; these are classified as safe for human consumption. So, concerns about toxic leachates while integration into food packing system are eliminated along with consumer acceptance (29). Unlike petroleum based plastic packages, these materials degrade naturally reducing microplastic pollution and landfill, enhancing the sustainability profile of the packaging. Consumer interpretability is one of the most significant advantages of using anthocyanin in food packing systems, as they are visually interpretable and consumers can make a quick and informed decision about the product freshness (30).

Despite having these advantages, these indicators face several challenges, which includes, stability, as anthocyanins are sensitive to temperature and light, leading to degradation or altered colour profiles (31) and environmental factors unrelated to microbial spoilage, like packaging atmosphere, may induce undesired changes in colour. Stabilization strategies are being explored to mitigate these issues (32).

1.5. CRISPR – Cas systems in Molecular Diagnostics:

Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR), along with Cas proteins, generate an adaptive antiviral defence mechanism in bacteria and archaea. When a bacterial cell is affected by a virus, a small piece of viral DNA is stored in the special section of bacterial DNA called the CRISPR array, which acts like a library of past infections. So, when

the same virus attacks them again, memory stored in this region is used to identify and destroy the virus. Here the CRISPR-associated Cas protein acts as a scissor to cut the viral DNA.

CRISPR systems are divided into two evolutionary classes, six types and more than 30 subtypes. While class I uses a group of proteins to form a Multiprotein Cascade and destroy the Viral DNA, Class II uses only one big protein called Multidomain effector nuclease (Eg: Cas9, Cas12, Cas13, Cas14) (33).

1.5.1. Why Class II enzymes became the workhorses of diagnostics?

Since class II systems rely on one polypeptide, it is easier to express, re-engineer, and purify them when compared to class I. This structural simplicity makes class II proteins a suitable candidate for diagnostic development and for genome editing. One of the critical limitations faced by well-studied class II nuclease Cas9, which is known for its sequence-specific double-stranded DNA cleavage, is that it will turn catalytically inactive after a single cleavage event for each target recognition, making it unsuitable for detecting nucleic acids that are present in low concentrations (34).

A pivotal advancement occurred when Cas13a, a Class II nuclease that targets RNA was characterized. Cas13a expressed a behavior that cleaves the target RNA and stays catalytically active, cleaving any nearby single-stranded RNA molecules. This phenomenon is called collateral cleavage activity. This mechanism can be used for signal amplification where recognition of a single target triggers hundreds or more secondary cleavages, making them a perfect molecular tool for biosensing platforms (35). Further functional screening of all the class II orthologs showed that Cas 12 and Cas14 also cleave the non-targeted single stranded DNA along with the targeted DNA, making them self-amplifying nucleases. This, when combined with reporter molecules that are either fluorescent or colorimetric, can be used for molecular diagnostics that are ultrasensitive (36).

This led to the development of diagnostic platforms based on CRISPR. Examples are SHERLOCK (Cas13a) and DETECTR (Cas12a) for detecting viral pathogens, mutations associated with cancer, and genes with antimicrobial resistance (37).

1.6. Novel mechanistic insights & enzyme engineering of Cas12a:

Cas12a offers several advantages that makes it an optimal choice for this study which includes, Trans cleavage activity that enables signal amplification in detection assays, without the need

for pre amplification of nucleic acid and operational flexibility. Also, it works with a single crRNA, unlike Cas9 which needs a tracrRNA along with crRNA (36).

A study showed that target specificity of Cas12a can be modulated by divalent metal ions, particularly by magnesium ions. CRISPR RNA-guided sequence recognizes the site of cleavage in double stranded DNA for Cas12a to cleave. A sequence of 5 to 7 nucleotides called the "seed region" is also a part of this mechanism, where mismatches can reduce the activity significantly. However, Cas12a exhibited greater tolerance for mismatches in this seed region when the concentration of Mg^{2+} ions is high; on the other hand, it exhibited higher specificity when Mg^{2+} concentrations decreased, resulting in reduced enzyme activity on mismatched targets. This discovery helps significantly for applications that need tuning of Cas12a specificity. For instance, in diagnostics for the detection of ultraprecise and specific variation in single nucleotide (lower Mg^{2+}), balanced Mg^{2+}concentrations can be used to reduce the off-target activity during gene editing, offering a nongenetic method to control CRISPR (38).

In the context of this project, this discovery can be useful while differentiating near-identical species of *Lactobacillus* without compromising the clear readout of the dye. The assay can be adjusted between "fine focus" and "wide angle" when detection at the genus level is adequate. This tuning can address off-target activity with almost no cost or regulatory complexity since magnesium salts are cheap and compatible with Cas12a reagents.

Another study on Cas12a trans cleavage activity showed that noncanonical duplexes of nucleic acid, like DNA: RNA hybrids, can also trigger the activity in addition to regular dsDNA. The efficacy of this activation varies by the duplex type as well as length (39). For this project, the hybrid activation mode mentioned in the study can link the output strictly with live cells. This strategy can be operationalized by designing a single stranded DNA probe of about 25-30 nucleotides, complementary to the mRNA viability marker that only living cells produce while making proteins. When this probe binds with mRNA, it forms an RNA-DNA duplex. To avoid other *Lactobacillus* species activating the system, SNP (Single Nucleotide Polymorphism, a single nucleotide difference that makes *L. agilis* unique) can be placed within the seed region. We can lyophilize these reagents on the 4D chip with one chamber containing Cas12a along with crRNA to target conserved DNA, reporters, and a moderate concentration of Mg^{2+} ions, while the other chamber is loaded with Cas12a with a strain-specific guide, DNA probe, and buffered with low Mg^{2+} (38), During the test, if the cells are dead, the chamber containing the DNA probe will remain colourless since dead cells lack an intact viability marker (mRNA) and

the other chamber gives colour whenever the DNA is present. This two-layer approach helps to screen the genus quickly and get live strain confirmation.

Similarly, in 2023, researchers developed a new technique called SAHARA (Split Activators for Highly Accessible RNA Analysis) where two single stranded DNA strands called activator strands are added. Each strand has two parts, where one part is the complementary to the RNA target and the other part contains the DNA sequence to mimic the double stranded DNA recognition site for Cas12a. When the RNA hybridizes with the activators, the two activator strands bind to its adjacent region bringing the DNA ends close, forming an RNA-DNA complex, resembling the double stranded DNA for Cas12a recognition. This can be programmed to detect different RNA by changing the crRNA and the activator strands, also it works at room temperature and can be multiplexed. In the context of this project, if *L. agilis* expresses species-specific transcripts, strain level resolution can be achieved by using SAHARA to detect RNA biomarkers (40).

1.6. Study Rationale and Proposed Approach

As stated earlier, *Lactobacillus agilis* a lesser-known probiotic species has emerged as a promising probiotic species. *L. agilis* is an anaerobe with effective carbohydrate metabolism and resistance to harsh gut conditions (11). Certain strains of *L. agilis* can adhere to intestinal mucosa and produce exopolysaccharides - features that enhance colonization and health efficacy (41). Despite having these attributes application of *L. agilis* as a probiotic is limited due to lack of rapid and cost-effective detection methods (42).

This study attempts to fill the gap in rapid and easily accessible detection methods for *L. agilis* by creating an integrated species level biosensing platform **(Figure 1)**. The method makes use of a colorimetric biosensor which responds to the metabolic activity of *L. agilis* by generating an instantaneous and visible signal, cutting out the need for sophisticated equipment. This detection mechanism is based on the observation that bacterial growth and its metabolic activities frequently made the surrounding medium more acidic, which lowers the pH locally (43). Notably the biosensor design leverages natural pH sensitive dyes like betalains (found in beetroot), curcumin (found in turmeric), and anthocyanins (found in red cabbage) as colour indicators. Additionally, the colour changes are reversible across physiological pH ranges making them a perfect candidate in this case (44) (45). By embedding these biocompatible dyes into a straightforward test system, the presence of *L. agilis* can be interpreted by a noticeable colour shift, offering a sustainable, user friendly and affordable detection method. These

biosensors have the potential to be used in real time for quality monitoring in food industries during packing, storage, and production.

This study further incorporates a CRISPR-Cas12a-based molecular confirmation phase to access and backup the biosensor's response specificity. Although high resolution strain level identification is frequently achieved by CRISPR systems, here Cas12a has been utilized for the verification of the species level presence of *L. agilis* by focusing on a genomic region that is conserved within the species but distinct from other closely related species of probiotics identified by comparative genomic analysis (46). This dual tier approach integrating metabolic detection with molecular confirmation ensures both biological specificity and visual accessibility (**Figure 1**).

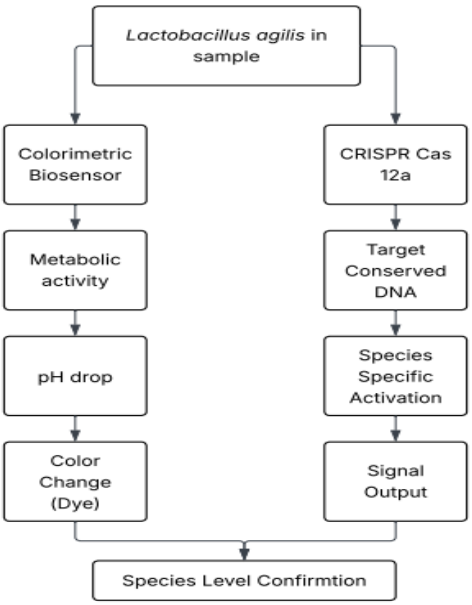

Figure 1 Schematic Overview of the *L. agilis* Detection Platform

As part of this work, the growth conditions for *L. agilis* are also optimized, including media formulation, anaerobic incubation, and quantification of growth phase using correlations between OD (Optical Density), pH, and CFU/mL. A reproduceable detection baseline for the biosensor system has been established by calibrating these variables. In parallel, molecular validation was started using published primers targeting a brief conserved region of *L. agilis*. Despite the successful amplification, direct sequencing is challenging due to the short fragment

length. Efforts concentrating on extending the amplicon or modifying sequencing techniques for accurate species level confirmation, constitute a crucial phase in molecular detection workflow.

In conclusion, this thesis investigates a novel *L. agilis* detection platform using CRISPR mediated species confirmation and biosensors using naturally derived dyes. This study aims to address the absence of rapid, affordable, and scalable detection technique in probiotic industry, in-order to encourage the incorporation of underrepresented probiotic species in food industry.

2. Aims and Hypotheses:

2.1. Standardization of *L. agilis*:

Aim:

To establish a relation between Optical Density (OD) and Colony Forming Units per Milliliter (CFU/ml), which will be reproducible under anaerobic growth conditions, for accurate estimation of viable cell counts during biosensor calibration.

Hypothesis:

Within a defined OD range, there is a linear correlation between OD and CFU/ml which can be used to estimate bacterial concentrations.

2.2. Development of Natural Dye based Colorimetric Biosensor:

Aim:

To formulate a bioink with pH sensitive natural dyes and to evaluate its capacity to visually represent metabolic activity of *L. agilis*.

Hypothesis:

Metabolic activity of *L. agilis* will result in the production of organic acids that reduce the local pH and trigger a detectable and noticeable color shift in the biosensor compared to uninoculated controls.

2.3. Examination of Colorimetric Reaction throughout Storage:

Aim:

To track pH and absorbance changes in dye infused bioink samples by comparing probiotic inoculated and control samples over a 7-day period at 4 °C to evaluate its response to *L. agilis* metabolic activity and the spoilage tendency of the bioink matrix.

Hypothesis:

Progressive decrease in pH and corresponding absorbance shifts will result due to metabolic acid production in probiotic inoculated samples. Control samples may also exhibit pH shifts overtime indicating the spontaneous spoilage.

2.4. Molecular Detection of *L. agilis* using CRISPR-Cas 12a:

Aim:

To design and validate the CRISPR-Cas 12a detection circuit for species specific confirmation of *L. agilis.*

Hypothesis:

When coupled with species specific crRNA targeting the Intergenic Spacer Region (ISR) in *L. agilis* species, CRISPR – Cas12a specifically recognize and cleaves the *L. agilis* DNA, discriminating them from the closely related *Lactobacillus* species.

2.5. PCR Amplification, Primer Development and Sequencing of the Targeted Region:

Aim:

To obtain an accurate sequence of *L. agilis* ISR region in the experimental strain for downstream crRNA design for CRISPR-Cas12a detection.

This will be obtained through:

1. Experimental validation of published primers for ISR amplification (47)
2. In silico primer design using Benchling for the extended region around the ISR, based on genome of *L. agilis* strain La3, to generate lengthy amplicon for improved sequencing accuracy and cross species comparison.
3. Whole Genome Sequencing (WGS) of the experimental *L. agilis* strain to confirm the sequence accuracy.

Hypothesis:

The published primers will yield a sequence by successfully amplifying the ISR region of *L. agilis*, while the extended primers will produce a longer amplicon with improved sequence reliability. The sequence finally obtained through these methods will be cross validated with WGS data to ensure specificity and suitability for crRNA design.

3. Materials

3.1. Colorimetric Sensor Development

3.1.1. Standardizing Viable Cell Concentrations of L. agilis

Reagents & Media:

1. MRS Broth Powder – Sigma-Aldrich
2. L-Cysteine (≥97% purity) – Aldrich Chemistry, Sigma-Aldrich
3. TWEEN® 80 – Sigma-Aldrich (Pcode: 102783372)
4. Deionized Water – Laboratory grade
5. MRS Agar – Sigma-Aldrich (Pcode: 102293132)
6. Anaerocult™ A gas-generating sachets – Merck, MilliporeSigma (Product No. 1.32381)

Biological Material:

Lactobacillus agilis – Experimental strain (−80 °C stock)

Equipment:

1. Laminar flow cabinet
2. Autoclave (120 °C, 15 min)
3. Incubator (37 °C, anaerobic)
4. Nanophotometer (Implen)
5. pH meter – SevenExcellence™ (Mettler Toledo, Singapore)
6. Precision wipes

3.1.2. Bioink Formulation and Dye-Based Colorimetric Assay

Bioink Materials:

Sodium alginate (Sigma-Aldrich, USA)

Cellulose nanocrystals (Sigma-Aldrich, USA)

Xanthan gum (Sigma-Aldrich, USA)

Whey Protein Isolate (WPI)

Dyes & Indicators:

1. Alizarin Red S – Sigma-Aldrich (1% aqueous solution)
2. Red cabbage – Fresh (400g) for anthocyanin extraction

Other Reagents:

1. Deionized Water
2. Phosphate Buffered Saline (PBS) – Sigma-Aldrich
3. MRS Agar - Sigma-Aldrich (Pcode: 102293132)
4. MRS Broth Powder - Sigma-Aldrich

Equipment:

1. Refrigerator (4 °C)
2. Freezer (−20 °C)
3. Hotplate & boiling setup
4. Micropipettes and sterile tips
5. Magnetic Stirrer
6. Hand blender

3.2. CRISPR-Cas12a Molecular Detection Circuit

3.2.1. Primer Procurement and PCR Amplification

Primers:

1. Agilis-F: TCGTAGCTTGCTACACCGATTG
2. Agilis-R: CATAATGACCATGCGATCATCA

Source: Synthesized by Sigma-Aldrich

PCR Reagents:

1. Taq DNA Polymerase 2X Master
2. Q5® High-Fidelity DNA Polymerase
3. dNTP Mix
4. Q5 Reaction Buffer
5. Nuclease-Free Water

Equipment:

1. Thermocycler - Thermo Fisher Scientific. (n.d.). *ProFlex PCR System*.
2. Nanodrop or Nanophotometer (Implen)
3. -20 °C freezer

3.2.2. PCR Product Validation and Sequencing

Gel Electrophoresis Reagents:

1. Agarose (2%) - Agarose, Genetic Analysis Grade – Fisher BioReagents, Fisher Scientific Ireland
2. 1x TAE Buffer
3. SYBR™ Safe DNA Gel Stain – Thermo Fisher Scientific (Cat. No. S33102)
4. 6X Gel Loading Dye (Purple) – New England Biolabs (Cat. No. B7024A)
5. HyperLadder™ 50 bp – Bioline (Cat. No. BIO-33054)

DNA Purification:

1. E.Z.N.A.® Gel Extraction Kit – Omega Bio-tek (USA)

Equipment:

1. Horizontal gel electrophoresis unit
2. iBright Imaging System – Thermo Fisher Scientific
3. UV transilluminator
4. Scalpel blades
5. Vortex, centrifuge

3.2.3. DNA Extraction for WGS and In Silico Design

DNA Extraction:

1. GenElute™ Bacterial Genomic DNA Kit – Sigma-Aldrich (Gram-positive protocol used)

Software & Databases:

1. NCBI BLAST (https://blast.ncbi.nlm.nih.gov)
2. Benchling (for in silico primer design)
3. FASTA genome files from NCBI GenBank (e.g., *L. agilis* La3, NZ_CP016766.1)

4. Methods:

4.1. Colorimetric Sensor Development:

This section outlines the experimental procedure for developing a colorimetric biosensor based on natural dyes for *L. agilis* detection.

4.1.1. Standardizing Viable Cell Concentrations of *L. agilis* Under Defined Growth Conditions

4.1.1.1.Media Preparation

MRS Broth was prepared by dissolving 52.5 g of MRS Broth Powder in 1L of deionized water, along with 0.05-0.1% L- cysteine (≥97% purity, Aldrich Chemistry, Sigma-Aldrich) and 1 mL of TWEEN® 80 (Sigma Aldrich, Pcode: 102783372). For solid media, 61.15 g of MRS Agar (Sigma Aldrich, Pcode: 102293132) is added to 1L of deionized water, along with 1ml of TWEEN® 80 and 0.05-0.1% L-cysteine and boiled to dissolve the media completely. Both the media are autoclaved at 120°C for 15 mins.

4.1.1.2.Bacterial Inoculation:

Under aseptic conditions in a laminar flow cabinet, a sterile pipette tip was used to transfer a small amount of *L. agilis* taken from -80°C stock culture into 50ml fresh MRS broth. The inoculated broth was incubated at 37°C under anaerobic conditions overnight to initiate bacterial growth.

4.1.1.3.Preparation of Defined OD samples:

A base dilution was prepared by mixing 100µl of overnight culture to 900 µl of MRS Broth (1:10 dilution), Nanophotometer (Implen) was used to measure the OD of this diluted sample. To prepare samples across target OD values ranging from 0.1 to 1.1, the OD of the base dilution was used to calculate the required volume of base dilution to mix with MRS broth to make up to 1ml using the formula $\mathbf{C_1V_1 = C_2V_2}$, where C1 is the OD of the base dilution, C2 is the desired OD and V2 is final volume (1ml). Although these values were predefined based on the calculated volume of each culture required for each target OD, the actual OD values measured showed a slight variation from the expected values **(Table 5).** Subsequently, 10 µL of each dilution was plated for CFU enumeration.

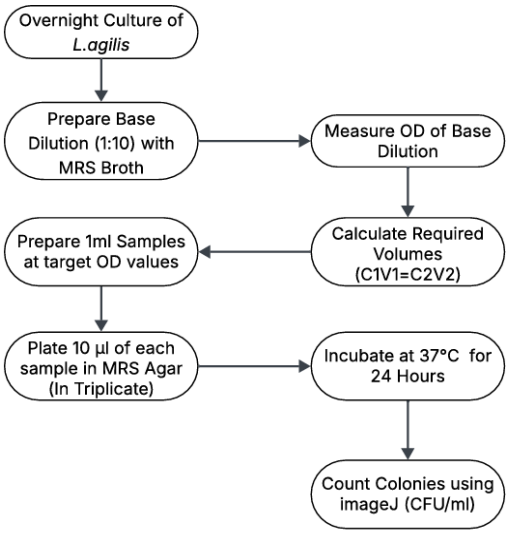

Figure 2 *Workflow for standardizing L. agilis viable cell concentration using OD and CFU enumeration*

4.1.1.4.Colony Counting:

Following plating, the plates were sealed inside sterile Ziplock bags along with Anaerocult™ A gas-generating sachets (Merck, MilliporeSigma, Product No. 1.32381) (activated by sterile water) and incubated at 37 °C for 24 hours. After incubation, the colony growth was photographed (**Figure 5)** and number of CFUs was quantified using ImageJ. **Figure 2** shows the entire workflow for standardization of *L. agilis*.

To establish a relation between OD and viable cell count, the average CFU/ml values for each sample was plotted against their corresponding OD. A linear regression curve was fitted to the data, giving a calibration equation to estimate the concentration of bacteria based on OD values

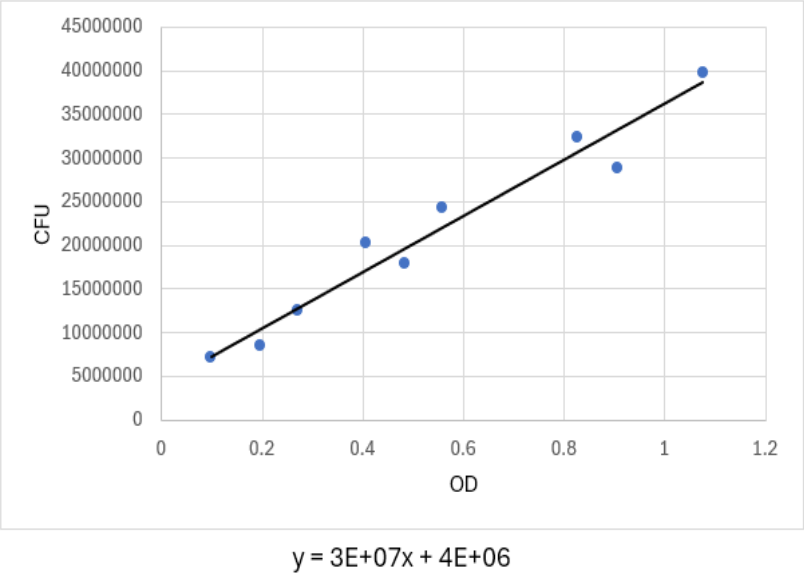

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Figure 4). This standard curve was subsequently used during bioink preparation to estimate the volume of probiotics to be added to the bioink at a standardized concentration of 10^7 CFU/mL (see section 4.1.2.3).

4.1.2. Bioink formulation and Dye based Colorimetric Assay:

Bioink was prepared following an internal, unpublished protocol developed by researchers in Agriculture and Biotechnology Research Centre, Plant and Agri biosciences, Aras de Brun, NUIG (48). The procedure was as follows

One gram of sodium alginate was dissolved in 100ml of distilled water using a hand blender for complete dispersion. Then, 1 g of cellulose nanocrystals and 0.5 g of xanthun gum are added and mixed using a magnetic stirrer for 10 minutes. Two grams of whey protein isolate was added to the mixture and stirred at medium speed for 1 hour in room temperature before transferring it to sealed container and storing overnight at 4°C to allow for full hydration. With this bioink, two different pH sensitive dyes were tested as indicators.

4.1.2.1. Alizarin:

In the first trail, 20ml of 1% alizarin solution, Alizarin Red S (from Sigma-Aldrich) was added with only 80 ml water to maintain the total volume of bioink as 100ml. However, this method caused non homogenous distribution of dye after refrigeration and excessively dark coloration due to higher volumes of dye. So, the protocol was adjusted by refrigerating the bioink overnight first and then added a reduced volume of 500 µL of 1% alizarin dye to 100 mL of fully hydrated bioink, ensuring better dispersion of the dye. It should be noted that alizarin is not approved as food additives or for direct incorporation into food under UK and EU food safety regulations (49) (50). Therefore, the data from alizarin based bioink can be used for

validating the sensor performance but are excluded from methods involving direct food incorporation and maybe they can be used in smart packing where indirect food contact is the goal.

4.1.2.2.Anthocyanin:

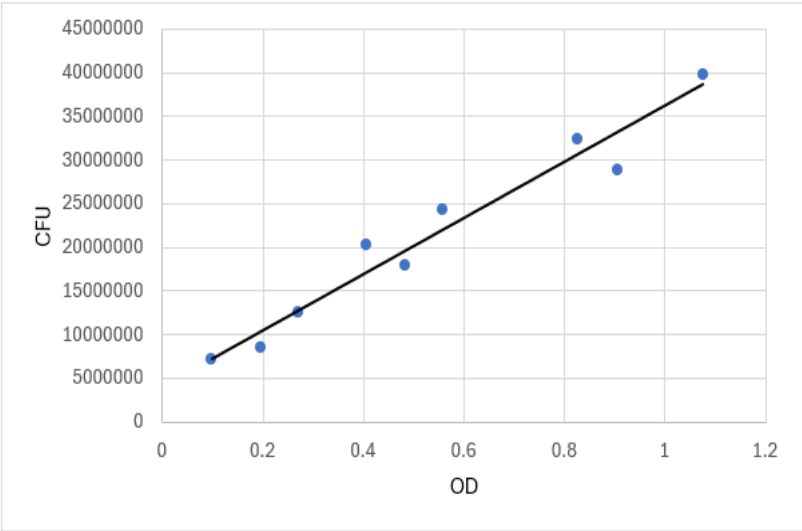
While waiting for the pure Anthocyanin powder to arrive, a natural extract was prepared by boiling 400g of chopped red cabbage in 1 liter of distilled water for 20 minutes, filtered and stored at -20 °C. During bioink preparation 100ml of red cabbage extract was directly used in the place of water in the bioink formulation.

4.1.2.3.Experimental Setup:

For both the dyes, two samples were prepared

1. Control: Bioink + Dye
2. Test: Bioink + Dye + Probiotic (*Lactobacillus Agilis)*

The probiotic concentration was standardized to 10^7 CFU/ml in compliance with the acceptable probiotic content in food. The OD vs CFU/mL calibration curve established earlier was used and the OD value corresponding to 10^7 CFU/mL was calculated using the linear equation derived from the graph to determine the volume of *L. agilis* containing MRS broth



y = 3E+07x + 4E+06

required (

Figure 4). Subsequently the bacterial culture was centrifuged at 1300 rpm for 3 minutes, the supernatant was discarded and the pellet was resuspended in 1ml of PBS before incorporation into the bioink prepared.

Also, a control experiment with a sample containing Bioink and probiotics without the indicator was performed to confirm that the results are due to probiotic activity and not because of intrinsic dye interactions. For this the sample is stored in 4°C. The CFU and pH measurements are made over a period of 5 days **(Table 6).**

4.1.2.4.pH, CFU and absorbance monitoring:

Over a 7day period, samples were stored in 4°C, and both absorbance and pH were measured to monitor the dye response and bacterial metabolic activity. Spectrophotometer was used for absorbance measurements. pH measurements were performed using SevenExcellence™ Multiparameter pH Meter (Mettler Toledo, Singapore). To prevent cross contamination between samples, the pH probe was rinsed thoroughly with distilled water and gently blotted dry with precision wipes. The probiotic inoculated samples were compared to control samples for colorimetric and pH shift measurements.

4.2. CRISPR-Cas12a Molecular Detection Circuit and Bioinformatics Workflow

4.2.1. Primer procurement and PCR amplification

4.2.1.1. Primer source:

Kim et al.'s published paper provided the primers employed in this study for the initial PCR amplification targeting *L. agilis* **(Table 1).** The primary purpose of these primers was to amplify a region conserved within the 16S-23S ribosomal RNA (rRNA) intergenic spacer, which facilitates the species level differentiation due to its conservation within the species and variability in sequence between closely related species.

Species	Target Region	Primer	Sequence (5′→3′)	Product Size (bp)
L. agilis	16S–23S ISR	Agilis-F	TCGTAGCTTGCTACACCGATTG	137
		Agilis-R	CATAATGACCATGCGATCATCA	

Table 1 Primer sequences published by Kim et al, that are used for PCR amplification of the 16S–23S ISR region in *Lactobacillus agilis*

4.2.1.2. Primer Design Methodology (Kim et al): (47)

1. Genome collection: The researchers utilized publicly accessible databases like NCBI GenBank and the Ribosomal Database Project (RDP) to generate a dataset of whole and draft genome sequences from a range of Lactic Acid Bacteria (LAB) species.

2. Target Selection: Considering the knowledge obtained from previous studies stating that the 16s-23s intergenic spacer region exhibits a significant degree of variation among species, while retaining conservation within the species, this region was chosen for designing highly species-specific primers.

3. **Multiple Sequence Alignment:** To find the species-specific conserved regions for primer design, the ISR (Intergenic Spacer Region) region sequences taken from the databases were manually aligned to guarantee the precision and specificity of the regions discovered.

4. **Primer Design:** Primers were designed to target the conserved region and are optimized to produce short amplicons suitable for quantitative PCR applications.

5. In Silico Validation: Primer specificity was confirmed by using in silico methods, particularly sequence similarity searches performed using NCBI BLAST, to ensure the primer's exclusive match to the targeted species and no cross-reactivity with other LAB species.

6. **Experimental Validation:** Further primers verified in silico were subjected to additional testing via PCR and qPCR using the genomic DNA isolated from different LAB strains. The experiment demonstrated the specificity of the chosen primers by confirming that they amplified exclusively the sequences from the intended LAB species.

4.2.2. Experimental Validation of Published Primers and Amplicon Sequencing:

4.2.2.1. Absence of sequence data from the literature:

The published study and its supplement did not report the precise nucleotide sequence of the amplicon generated by the forward and reverse primers even though those primers had undergone extensive specificity validation. Since, the exact sequence is essential especially while developing crRNAs for downstream CRISPR-Cas12a detection tests, PCR followed by sequencing are used to identify the area amplified.

4.2.2.2. Primer Synthesis

After being taken from the published paper, the forward and reverse primers were commercially synthesized. The lyophilized oligonucleotides at standard purity were provided by Sigma

Aldrich. Upon delivery the primers were reconstituted in nuclease free water to a stock concentration of 100 µM (micromolar) and stored at −20 °C for future use.

4.2.3. PCR Amplification of the Targeted Intergenic Spacer Region in *L. agilis*:

4.2.3.1. Genomic DNA Extraction:

Using the GenElute™ Bacterial Genomic DNA Kit (Sigma-Aldrich, USA) and in accordance with the manufacturer's instructions DNA was extracted from *Lactobacillus Agilis* cultures (51). Since *L. agilis* is gram positive, the gram-positive protocol was selected and followed as outlined in the kit protocol for gram positive bacteria. The extracted DNA was eluted in nuclease free water, quantified using nanodrop and stored at −20 °C.

4.2.3.2. PCR Reaction Setup:

The conserved ISR region between the 16S and 23S rRNA genes of *L. agilis* was amplified using PCR by species specific primers outlined in Kim et al. To validate the basic amplification of the target region and the performance of the primer, initial PCR amplifications were conducted using standard Taq DNA Polymerase due to its robustness and ease of use during early optimization. To confirm precise amplification with less nucleotide misincorporation, the experiment was repeated using Q5 high fidelity DNA Polymerase.

The detailed PCR reaction components for both Taq and Q5 polymerase are summarized in **Table 2** and the thermal cycling conditions used for amplification are presented in **Table 3.** Taq polymerase was used with its 2x Master Mix, which contains buffer, dNTPs and Mgcl2 in a single solution. While Q5 DNA Polymerase requires individual addition of each component, including separate Q5 buffer and dNTPs.

Parameter	**Taq DNA Polymerase µl**	**Q5 high fidelity polymerase µl**
dNTPs		1
Forward Primers	1	2.5
Reverse Primers	1	2.5
Polymerase	12.5	0.5
DNA Template	4	8
Nuclease Free water	6.5	25.5
Q5 Buffer		10
Total Volume	25	50

Table 2 : PCR Reaction Components per tube for Taq and Q5 Polymerase for Amplification using Published primers.

Step	Temperature (°C)		Time		No of Cycles
	Taq	Q5	Taq	Q5	
Initial Denaturation	94	98	10 mins	2 mins	1
Denaturation	94	98	30 secs	20 secs	For Taq - 30 For Q5 - 35
Annealing	60	63	30 secs	30 secs	
Extension	72	72	30 secs	30 secs	
Final Extension	72	72	5 mins	5 mins	1
Hold	12	12	∞	∞	

Table 3 : Thermal Cycle Conditions Used for PCR Amplification using published primers

4.2.3.3. Initial Confirmation of PCR Amplification by Agarose Gel Electrophoresis:

To access the success and specificity of amplification, the PCR products were analyzed using agarose gel electrophoresis, which was performed using standard molecular biology protocols. A 2% agarose gel was prepared using 1x TAE (Tris-acetate-EDTA) buffer and stained with 5 µl of SYBR™ Safe DNA Gel Stain (Thermo Fisher Scientific, Cat. No. S33102) according to the manufacturer's instructions. Each PCR product was mixed with 4 µl of 6X Gel Loading Dye (Purple) (New England Biolabs, Cat. No. B7024A) and loaded into individual wells in the gel. Electrophoresis was carried out alongside a 50 bp DNA ladder (HyperLadder™ 50 bp, Bioline, Cat. No. BIO-33054) at 45-50 volts for 1 hour and visualized under iBright Imaging System (Thermo Fisher Scientific, USA) (**Figure 9)**

4.2.3.4. Amplicon Purification and Gel Validation Prior to Sequencing:

The gel was placed under UV transilluminator for clear visualization of DNA bands by eye, followed by careful excision of the desired amplicon from the gel using a sterile scalpel blade. Purification of DNA from the excised gel slice is done using E.Z.N.A.® Gel Extraction Kit (Omega Bio-tek, USA) following the manufacturer's protocol (52).

After purification the DNA was eluted in a final volume of 35 µl, further the concentration of the DNA was measured using Nanophotometer and found to be approximately 15 ng/µL. Since most commercial companies need a minimum of at least 10 ng/µL for short sequences, this step is to ensure that the purified amplicon was present in adequate concentration for successful submission to sanger sequencing. 8 µL of the sample was reloaded on a 2% agarose gel to confirm the integrity of the eluted DNA. A faint band was observed, which is corresponding to 137bp confirming the presence of the correct product **(Figure 11)**. The remaining 27 µL of sample is stored at -20 °C for downstream applications.

For sequencing, 10 µL of the sample will be transferred to two separate tubes, one containing the forward primer and another with reverse primer and will be submitted for sequencing.

4.2.4. Strategies for Obtaining the Full Sequence of the Amplified Region:

We need the complete and accurate sequence of the amplified region for downstream application like CRISPR Cas12a crRNA design. Even though the 137 amplicon was successfully generated and submitted for sequencing, sequencing error may appear with these kinds of short amplicons, specifically near the primer binding region because of poor peak resolution. So, following the successful amplification of the 137 bp amplicon, two parallel strategies were considered to achieve this goal of obtaining reliable target sequence, each with its own advantage and limitations.

Strategy 1: Whole Genome Sequencing (WGS) of *L. agilis* Strain:

The most thorough and precise method is the WGS of *L. agilis* strain used for experimental validation. Upon sequencing the raw reads will be obtained in FASTQ format, and will undergo preprocessing, including trimming and quality filtering. Reads will be assembled using reference-based genome assembly, where high quality reads are aligned to a closely related reference genome using software tools. Following assembly, the 16s and 23s rRNA genes will be identified in the draft genome. The ISR region located in between these rRNA genes will be identified and will be aligned against multiple *L. agilis* strain to confirm sequence conservation. To ensure the specificity of the assay the ISR region will be BLAST against other closely related *Lactobacillus* species.

This method ensures that the sequence obtained is reliable, but this is resource intensive in addition to high cost. This constrained its implementation during the current stage of the project. Therefore, as an interim measure, the second strategy was adopted. Nonetheless, this method is planned for implementation in future stages of the research.

Strategy 2: In Silico Analysis Based on a Closely Related Reference Strain

To find a very similar reference genome for primer construction, Kim et al's forward and reverse primer sequence was submitted to NCBI BLASTn. For in silico analysis, several strains of *Lactobacillus agilis* that showed high sequence similarity to the query was downloaded in FASTA format. The genome of *L. agilis* strain La3 (RefSeq: NZ_CP016766.1) was chosen among the top BLAST hits. Using the published primers, in silico PCR was performed on La3 genome to locate the ISR region and it showed the expected 137 amplicon. With that, new

primers are designed to flank a larger segment surrounding the ISR region in La3. These extended primers will then be applied to the experimental strain, with PCR performed under the standard amplification conditions used in this study, to determine if the genomic region is amplified at the expected length, acknowledging that amplification success can be affected by sequence variation between La3 and experimental strain. If the correct-sized product is obtained, the longer amplicon will be submitted for sequencing in addition to the previously obtained 137 bp amplicon, enabling more accurate characterization of the target region. Prior to designing the extended primers, the obtained 137 bp from in silico PCR of La3 was evaluated for sequence specificity using BLASTn searches (Restricted, Unresticted, Exclusion methods) (See section 5.2.4).

Table 7 summarizes the extended primer sequences, their source strain and predicted amplicon length; **Table 4** shows the PCR conditions for the extended primers using Taq polymerase. **Figure 12** shows the location of this target region on the circular genome map, while **Figure 13** displays linear annotated view of the region sequences and primer binding sites. These results will be cross validated with both the previously amplified 137bp sequence (See section 4.2.3.4) and with whole genome sequencing data of the experimental strain (See section 4.2.4, Strategy 1). **Figure 3** illustrates the entire workflow for developing the *L. agilis* specific crRNAs for CRISPR – Cas12a circuit.

Step	**Temperature**	**Time**	**No.of.Cycles**
Initial Denaturation	95	5 mins	1X
Denaturation	95	30 sec	40X
Annealing	61	30 sec	
Extension	72	1 min	
Final Extension	72	10mins	1X
Hold	12	∞	

***Table 4*:** PCR Operating Conditions for Primers Designed Using Benchling

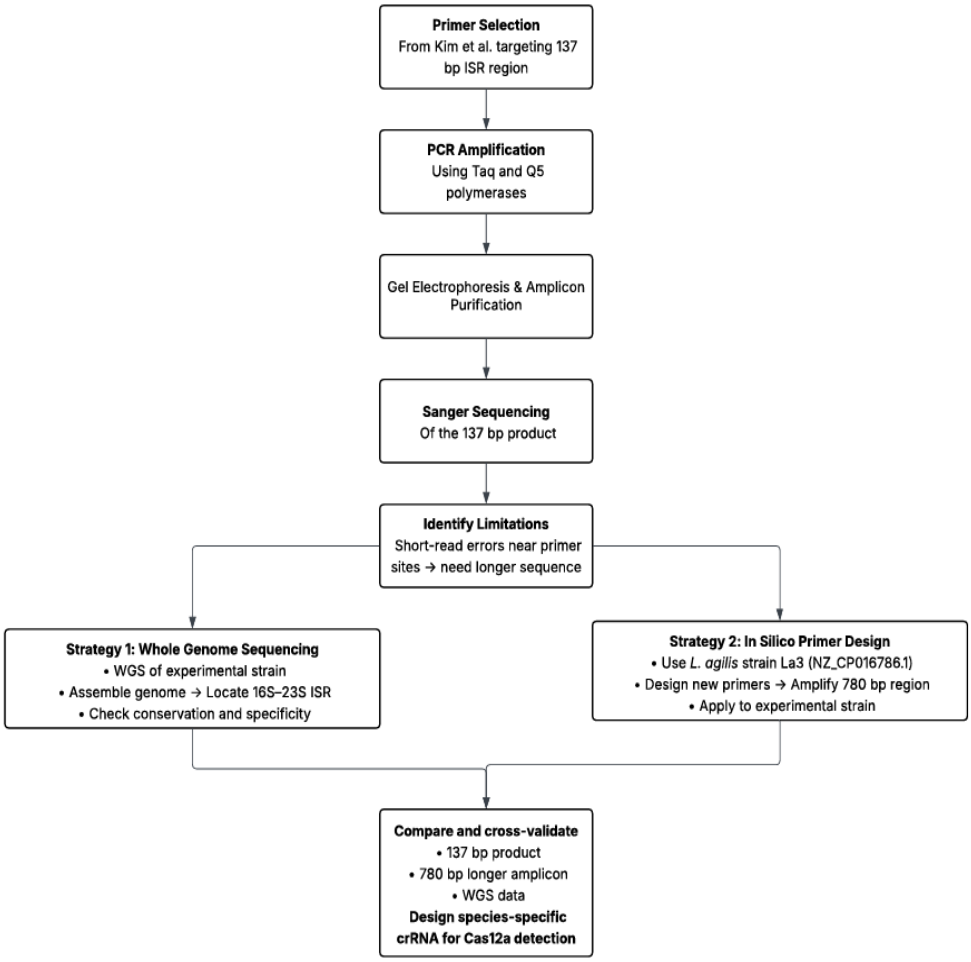


Figure 3: Workflow for Designing Species Specific crRNA

5. Results:

5.1. Colorimetric Sensor Development

5.1.1. Standardizing Viable Cell Concentrations of *L. agilis*

To develop a standard curve for the estimation of bacterial concentration based on OD, CFU/ml was calculated for a range of optical densities (0.1-1.1) by plating serial dilutions of the cultures. Each dilution plated in triplicate on MRS agar. **Figure 5** shows a representative CFU growth plate image from one dilution point, showing the morphology of colonies observed. CFU/ml was calculated for each sample and plotted against their corresponding OD; the corresponding values are summarized in **Table 5**. As shown in **Figure 4** a linear relation was observed between OD and CFU, confirming that OD can be used to estimate viable cell counts within this range.

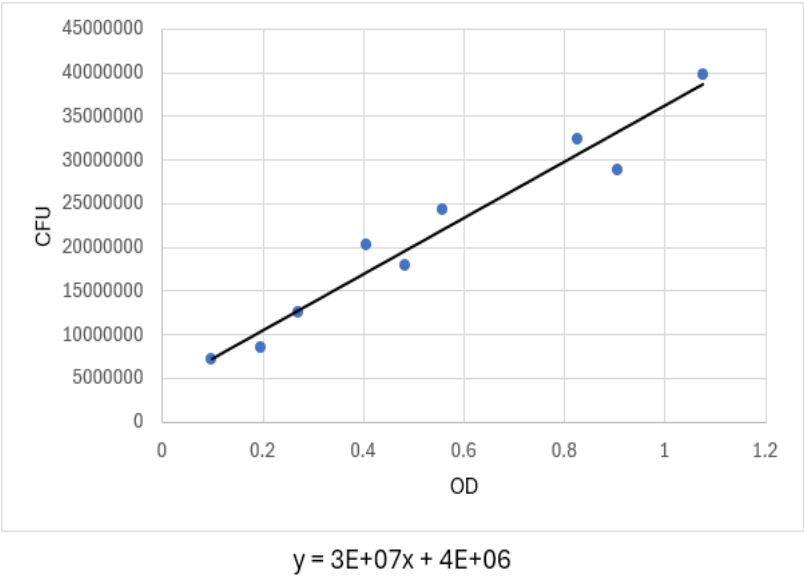


Figure 4: Calibration curve showing a linear relation between OD and CFU

Figure 5: Picture of colonies on MRS agar with OD:1.075

	Expected OD	Volume from overnight culture (µL)	To add Volume of fresh MRS (µL)	Measured OD	COLONY COUNT 3times dilution in base 10	Average cfu	Scientific number
1	0.1	13	988	0.098	71,74	7200000	7E+6
2	0.2	25	975	0.197	95, 75	8500000	8 E+6
3	0.3	38	963	0.271	133, 116	12500000	1 E+7
4	0.4	50	950	0.407	234,170	20200000	2 E+7
5	0.5	63	937	0.483	163,198	18000000	2 E+7
6	0.6	75	925	0.557	250,236	24300000	2 E+7
7	0.8	100	900	0.826	442,206	32400000	3 E+7
8	1.0	125	875	0.906	293,282	28800000	3 E+7
9	1.1	138	862	1.075	371,423	39700000	4 E+7

Table 5*:* Summary of OD measurements, dilution volumes, and corresponding average CFU counts used to generate the standard curve

5.1.2. Dye Based Colorimetric Response

5.1.2.1.Visual Color Change and pH Shift:

Over the course of the 7 days, bioink samples made using Anthocyanin as an indicator, displayed a noticeable change in colour. Probiotic inoculated samples showed a more noticeable and visible change in hue when compared to control samples. This corresponds with pH gradually dropping overtime.

Bar graph comparing the pH values of samples with and without probiotics (**Figure 6**), show a noticeable divergence starting day 2. Probiotic samples reached a minimum pH by Day 5, reflecting the production of acid due to bacterial metabolism. The control samples without the probiotic exhibited a gradual decline in pH, reaching below pH 6 by day 4-5. Factors like dye

degradation, bioink components degradation may have contributed to this acidification in the absence of probiotics.

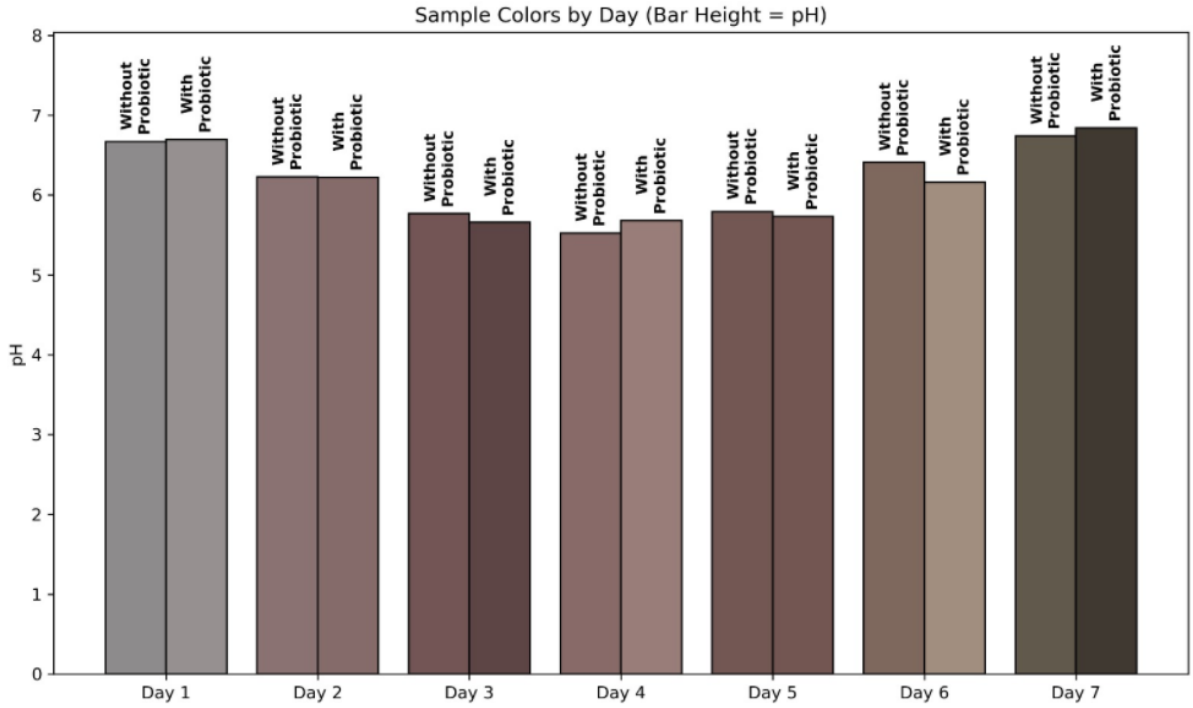


Figure 6 : Visual and pH Changes in Anthocyanin-Based Bioink with and Without L. agilis During 7-Day Storage at 4 °C

<u>Spectrophotometric Absorbance Monitoring</u>

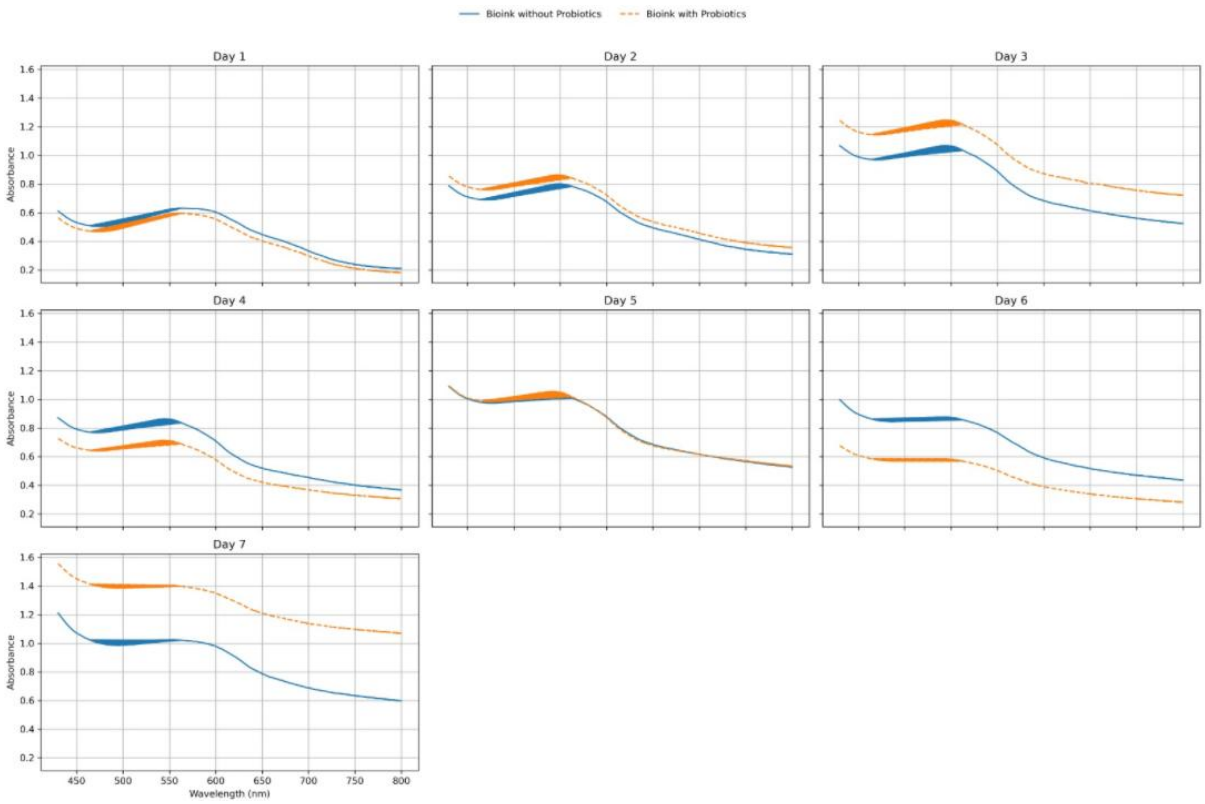


Figure 7: UV–Vis Spectral Changes of Anthocyanin-Based Bioink with and Without L. agilis Over 7 Days

The observed color transitions are further confirmed by UV-Vis absorbance spectra **(Figure 7).** At first, the spectral profiles of the probiotic and control samples were similar with peak absorbance (Purple) in 550-580 nm range. However, by Day 3 probiotic samples exhibited decreased absorbance at higher wavelengths and a blue shift in peak location, which is consistent with structural changes bought on by acid in Anthocyanin molecules (see section 6.2). By day 7, a noticeable reddish-brown color was observed in probiotic samples, with greater absorbance at lower wavelength (~430–450 nm). For the control samples the spectral forms remain largely constant.

<u>pH and Bacterial Growth Correlation:</u>

Direct measurements of CFU/mL and pH over a 5-day period in a sample (Bioink + Probiotic) without any indicators, clearly demonstrate an inverse connection. pH decreases from 6.8 to 6.0, as CFU increased from 1.7×10^{6} to 8.5×10^{7} **(Table 6)**. This is consistent with the known knowledge about the *L. agilis* metabolic activity which produces acid as a by-product. However, during day 4, despite the drop in pH, CFU also started to decline. This may indicate bacterial growth inhibition due to acid stress or nutritional depletion **(Figure 8).**

Day	pH	CFU/ml
1	6.802	1.70E+06
2	6.326	3.50E+07
3	6.043	8.50E+07
4	5.864	3.80E+07
5	5.735	1.90E+07

Table 6 : pH and CFU/mL measurements over a 5-day period for bioink containing Lactobacillus agilis without indicators

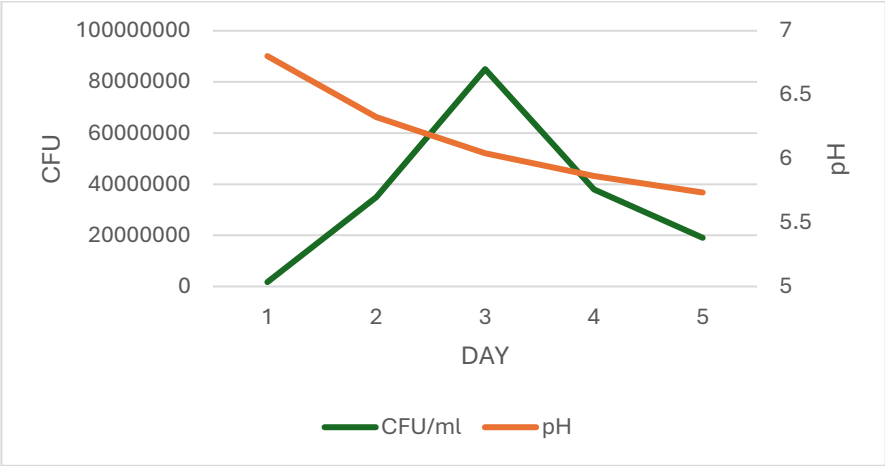


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Figure 8 : Relationship between pH and CFU/mL over a 5-day period in bioink containing Lactobacillus agilis without indicators

5.2. CRISPR

5.2.1. PCR Amplification Results

PCR results using Kim et al's primers showed a clear 137bp band on the gel **(Figure 9)** using both Q5 and Taq polymerase with one repeat for each. The Q5 polymerase amplified product appeared more intense likely due to high reaction volume.

Figure (10) shows amplification using the in silico designed primers from strain La3 produced a 780bp band and the smear is likely due to the high reaction concentration.

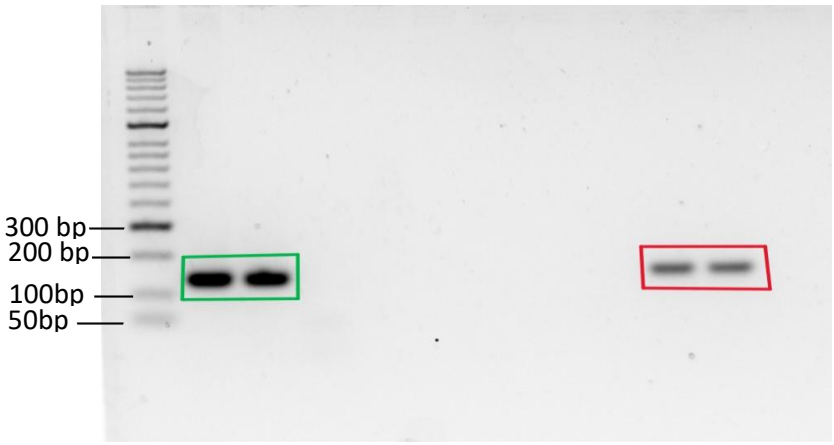


Figure 9: Agarose gel showing 137 bp amplicon (Lane 1: DNA ladder-50bp, Bands marked with green box – obtained with Q5 polymerase, Bands marked with red box – obtained with Taq polymerase)

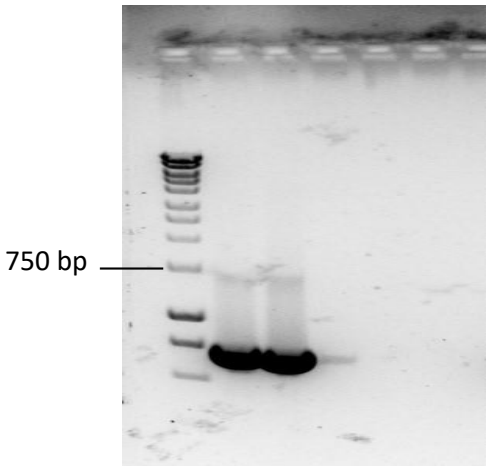


Figure 10: 780 bp amplicon generated using benching designed primers (Lane 1: 100bp DNA ladder, Lane 2 and 3: ~ 780 bp band produced by benchling designed primers)

5.2.2. Sanger Sequencing

Figure (11) shows a band corresponding to the expected size, obtained through agarose gel electrophoresis of the purified 137 bp amplicon after DNA extraction **(Figure (9)).** This confirms the sample integrity for Sanger sequencing.

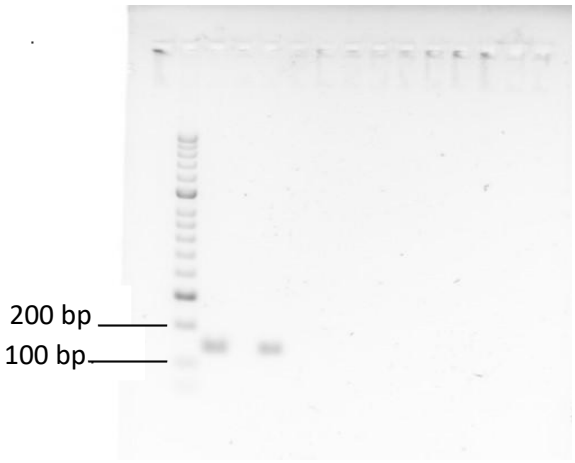


***Figure 11**:* 137 bp amplicon loaded post DNA extraction (Lane 1: DNA ladder-50bp)

5.2.3. In Silico Primer Design

Using Benchling, primers were designed to flank the ISR region in the La3 reference genome. **Figure (12)** shows the location of this target region on the circular genome map, while **Figure (13)** displays linear annotated view of the region sequences and primer binding sites. **Table 7** summarizes the primer sequences and their source strain. These primers were used in PCR and produced the expected amplicon as shown in **Figure (10).**

***Figure 12**:* Circular genome map showing ISR region and primer locations.

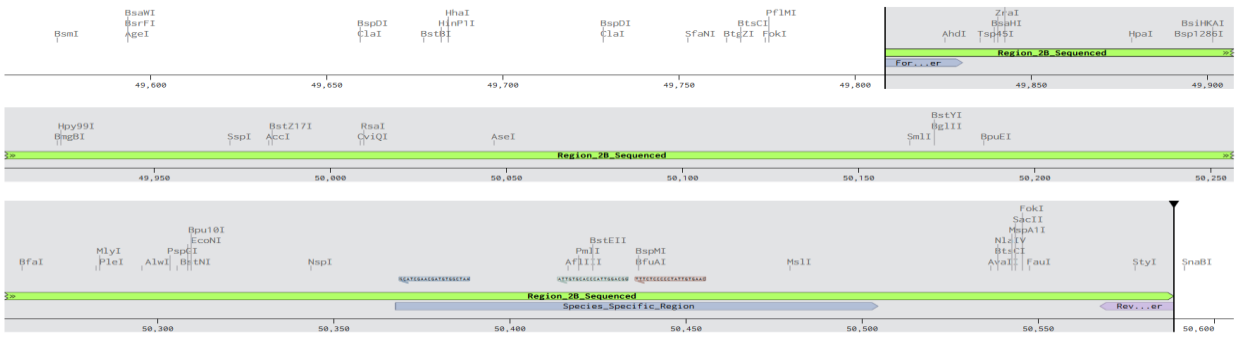


***Figure 13**:* Linear view of ISR region with designed primers

Strain	Forward Primer	Reverse Primer	Amplicon Size
La3 (RefSeq: NZ_CP016766.1)	TGTTATTAACTGACGCACCGTC	TCATTACCTTGGTAGGCCGTT	780 bp

***Table** 7:* Primer sequences and predicted amplicon size from La3

5.2.4. In Silico Validation of *Lactobacillus agilis* ISR-137 Amplicon Specificity

As WGS data for the experimental strain was not available at this time, a closely comparable reference genome was utilized to determine species specificity. The complete genome of *L. agilis* LA3 strain was used as a template for in silico PCR , with published primers that target the *L. agilis* ISR region. The result yielded the 137bp amplicon **Figure 14**, with which three BLASTn searches were performed to access the species specificity (**Table 8**).

1. A Restricted Search (NCBInt, Taxid: 1601 only), yielding a perfect match to strains of *L. agilis*, including 16s RNA sequence from variety of isolates and complete genomes all with 100% query coverage and 99-100% identity
2. Unrestricted search: High scoring matches to sequences identified as *Limosilactobacillus* fermentum and unclassified lactobacillus species are obtained. These sequences reflect highly conserved rRNA regions in closely related taxa, showing phylogenetic proximity.
3. Exclusion search: The absence of named non agilis species with high identity matches was confirmed by this search with unrelated genera like Bifidobacterium and L. fermentum yielding the closest results, with weaker E values despite high percentage identity. This shows phylogenetic proximity.

Together these findings show that although the ISR locus (obtained through in silico PCR of La3 genome), is highly conserved among the *L. agilis* strains, it shares sequence similarity with some closely related lactobacilli, but no significant matches in other ligilactobacillus species, indicating its suitability for *L. agilis* specific CRISPR CAS12a design

Search type	Amplicon	Strain / Species	Accession	% Identity	Query Cover	E-value
Restricted (taxid:1601)	ISR-137	L. agilis La3 (complete genome)	CP016766.1	100.0	100%	6e-68
	ISR-137	L. agilis AM_LB8 (complete genome)	CP104397.1	100.0	100%	6e-68
	ISR-137	L. agilis strain 1472 (partial 16S)	MT573856.1	99.27	100%	3e-66
	ISR-137	L. agilis strain 1420 (partial 16S)	MT573828.1	99.27	100%	3e-66
Unrestricted	ISR-137	Limosilactobacillus fermentum strain NMCC-20 (partial 16S)	MK615937.1	99.27	100%	4e-61
	ISR-137	Lactobacillus sp. UVAS:RY12 (partial 16S)	MG938656.1	99.27	100%	4e-61
Excluded *L. agilis*	ISR-137	Limosilactobacillus fermentum strain NMCC-19 (partial 16S)	MK614258.1	99.27	100%	4e-61
	ISR-137	Bifidobacterium longum strain YGMC0602 (chromosome)	CP137051.1	99.27	100%	4e-61

Table 8: BLASTn search results for the 137 bp ISR amplicon (Obtained through in silico PCR of La3).

5′-TCGTAGCTTGCTACACCGATTGAAAATTGAGTGGCGAACGGGTGAGTAACACGTGGGTA

ACCTGCCCAAAAGAGGGGGATAACACTTGGAAACAGGTGCTAATACCGCATAACCATGAT

GATCGCATGGTCATTATG-3′

Figure 14: 137 Nucleotide Sequence Obtained By In Silico PCR using published primers in La3 whole genome.

6. Discussion:

6.1. Implications of OD–CFU Calibration for Assay Reproducibility

To ensure assay reproducibility and interpret sensor outputs, establishing a reproducible relationship between OD and CFU is essential. In this study, the OD and CFU/ml calibration curve for *L. agilis* was generated under defined conditions in MRS media, which showed a linear relation in the OD range of 0.1-1.1**(Figure 4).** This confirms that OD measurements when conducted under controlled growth conditions can reliably estimate viable bacterial concentrations and will be helpful while preparing bioink at 10^7 CFU/mL, recommended dose of probiotic in dairy (53) (54) (55), without CFU plating for each experiment.

This result is consistent with earlier research on lactic acid bacteria including *L. agilis*, especially during the early to mid exponential phases. For instance, Stevenson et al., 2016 (56) established methods and equations to convert OD reading to CFU/ml and validated it across different bacterial species. In this study we deliberately limited our OD range to avoid recognized issues that might skew linearity above OD of 1.0 in dense cultures, such as light scattering and cell aggregation.

These calibration curves enable more robust comparisons among experimental batches and serve as a basis for standardizing input concentrations. They improve data interpretability when accessing dose response and storage stability. In the context of *L. agilis*, whose growth may slightly differ from other LAB species, having a species-specific calibration is critical. Some strains may be motile and exhibit altered biofilm formation, affecting the measurements if not accounted for (10). Defined growth conditions, consistent media composition, and defined OD sampling helped mitigate these problems. Without this baseline curve linking biomass with viable cell counts, interpretations of biosensor responses downstream lack biological context. Thus, the OD–CFU relation established here not only gives the growth kinetics of *L. agilis* under controlled conditions but also provides a foundation upon which further biosensor design can be built confidently.

6.2. Bioink-Based Colorimetric Detection: Efficacy and Interpretation

The anthocyanin-based bioink created in this study, effectively displayed a distinct and responsive pH-sensitive colorimetric shift in the presence of probiotic bacterial activity. These findings provide credibility to the possible application of naturally derived food-grade indicators in the creation of biosensors for smart food packaging and probiotic monitoring.

The noticeable colour shift that was seen in the probiotic samples (**Figure 6**) with anthocyanin can be explained by the well-established pH-dependent structural changes of anthocyanin. Anthocyanins are a type of flavonoid commonly found in fruits and vegetables and are known to vary structurally, depending on the pH of the surrounding environment. Anthocyanins predominantly exist in the quinoidal base form at a near neutral pH of 6.5-7, which absorbs in the 550-580 nm range and gives off a purple tint. The molecules change into flavylium cation form when the pH drops, in our case pH drop due to acidification by probiotic metabolism. This form absorbs shorter wavelengths of about ~430–450 nm, resulting in reddish-brown hue (57). The absorbance data from day 3 onwards showed a distinct transition in spectrum, consistent with the medium getting acidic as a result of *L. agilis* metabolism **(Figure 7).**

This structural change is clearly demonstrated by the blue shift in the UV-Vis absorbance peak from higher to lower wavelengths **(Figure 7).** The transition to the higher energy state of absorbance represents a change in electronic structure resulting from increased protonation under acidic conditions, which is represented by a blue shift. These results are consistent with earlier studies using anthocyanins as a natural pH indicator because of their colorimetric and spectral sensitivity to acidic conditions (58) (59).

To confirm that the results are due to probiotic activity and not dye interactions, a control experiment without indicator (Bioink + Probiotic) was performed. The resulting pH vs CFU graph **(Figure 8),** showed a classical microbial growth pattern, where the CFU/ml rose sharply from day 1 (1.7×10^6) to day 3 (8.5×10^7), indicating the exponential growth phase of *L. agilis*, with simultaneous drop in pH from 6.8 to 5.7 confirming acid production because of bacterial metabolic activity. Decline in CFU/ml after day 3 despite the continued decrease in pH, suggests onset of bacterial death phase, which may be the result of nutrient exhaustion or acid inhibition of key metabolic pathways.

The trends observed during the exponential growth of *L. agilis* from day 1 to day 3 and its decline through day 5, provides a baseline for interpreting the behaviour of anthocyanin-based probiotic samples. Even though direct pH measurements and CFU enumerations are only carried out over a 5-day period in bioink samples without indicators, similar acidification patterns were seen in trails with anthocyanin via both pH measurements and spectral changes, allowing logical correlation. The consistent onset of colour and spectral shifts from day 2 aligned with metabolic activity window confirmed in indicator free trail.

While the control samples are free of probiotics, a gradual decrease in pH was still observed **(Figure 8).** This may be because of the dye degradation or CO2 absorption forming carbonic acid. However, there are no major changes in control sample's colour or spectral stability, suggesting that anthocyanin can remain relatively stable in refrigerated conditions (4 °C). This property is very useful when it comes to smart food packing where it's crucial to ensure that the changes in signal is primarily due to biological triggers.

For real world applications, further investigations are needed to assess the stability of anthocyanin based bioink, in addition to validating the response specificity of anthocyanins against other spoilage organisms in order to evaluate its specificity to probiotics versus general acidification.

6.3. CRISPR-Cas12a Molecular Confirmation Assay

A major challenge in probiotic diagnostics is identifying the target organism precisely in the background of phylogenetically identical species, specially for species with limited taxonomic infrastructure like *L. agilis*. This study uses CRISPR-Cas12a system to provide species level molecular confirmation for *L. agilis* by precisely targeting the 16S-23S intergenic spacer region, which gives higher differential resolution than conserved ribosomal genes alone.

To facilitate the molecular detection of *L. agilis*, PCR amplification was performed using two primer sets: One is a pair of published primers targeting a short 137 bp region unique to the *L. agilis* species and a second pair of new in silico designed primers intended to amplify a longer fragment (~780 bp) flanking the 16S–23S intergenic spacer (ISR) region.

Initial PCR using the published primers yielded a 137 bp amplicon confirming primer functionality and the presence of ISR region in the *L. agilis* strain under investigation **(Figure 9)**. This aligns with the original goal of validating the species level identity of the experimental strain for downstream CRISPR applications. The published literature did not provide the sequence of the 137-amplicon necessitating further sequencing to access its suitability for CRISPR crRNA design.

Despite the known challenges associated with sequencing short amplicons – particularly in regions near the primer binding sites due to overlapping peaks of chromatogram, sanger sequencing for the 137bp was pursued. The successful visualization of the purified amplicon post DNA extraction from the gel indicates successful DNA recovery, DNA integrity and adequate concentration of DNA for sequencing of the 137bp amplicon **(Figure 11).** This

experimental setup bridges the gap left by the literature, which lacks published sequences for the 137bp ISR amplicon and enabled direct molecular characterization of the region for CRISPR assay.

To overcome the limitations of using short sequences during sequencing, a complimentary strategy was implemented, in which primers were designed for regions flanking the ISR in *L. agilis* strain La3 to amplify a longer ~780 bp region using Benchling platform **(Figure 12, Table 7)**. The band observed corresponding to the expected size **(Figure 10)** confirms the correctness of the primer designed and the conservation of this extended ISR region across strains. The main purpose for designing this extended fragment is to improve sequencing resolution (60). In Figure 10, the presence of discrete, high intensity bands shows low off target amplification and high target specificity, which is crucial for downstream CRISPR-Cas12a applications. Although the minor smearing observed is likely due to elevated template or primer concentrations, which can be optimized in the future, the primers yielded a strong band migrating close 750bp ladder. While the sequencing results are not available at the time of writing, this strategy reflects a targeted effort to strengthen molecular confirmation of species. This represents an important step towards ensuring the CRISPR – Cas12a specificity and highlights the importance of iterative sequencing in resolving taxonomic ambiguity.

This work demonstrates the possibility to integrate wet lab and computational workflows into an iterative validation loop. This method is very helpful while working with species like *L. agilis* that are not well understood. Also, crRNAs can be rapidly redesigned and revalidated whenever new sequence data becomes available, making CRISPR systems a perfect candidate.

6.4. Comparative Context

The combination of dye based visual readout along with CRISPR mediated molecular confirmation ensures both functional and taxonomic validation of *L. agilis*. This biosensor has a potential for integration into smart packaging systems and offers a low cost, point of use applicability.

Bull et al. (2013) highlighted the limitations of metagenomic approaches in LAB such as *L. agilis*, noting that they are often missed in sequencing datasets (61). This may be due to several factors like their low abundance, making them undetected in sequencing reads that prioritize dominant species or due to the lack of well annotated reference genomes in datasets resulting in misclassification or total omission during profiling. These factors underscore the need for alternative species-specific detection methods. This study directly addresses this limitation by

using CRISPR – Cas12a system for precise molecular identification of *L. agilis* without the need for sequencing or exhaustive database annotations. As such this method complements metagenomic tools by filling a critical gap between probiotic species level validation and detection.

Importantly, multiplexing and adaptation are made possible by the CRISPR detection circuit. The same platform can be modified to identify different probiotic species or even genes linked to antibiotic resistance by altering the crRNA sequence.

6.5. Limitations of the Study

There are several limitations to this study, despite having promising outcomes. While the OD-CFU correlation was strong, due to matrix interference or clumping of cells, variability may arise, especially in real world samples. The dye-based detection system can effectively capture the acidification trends, but in long term its stability was limited, as we can observe a pH shift in control samples (**Figure 6**), posing challenges in extended shelf-life applications and demanding stabilizing strategies.

Although ISR target was successfully amplified and verified, the core functionality of the CRISPR system i.e., target cleavage and triggering detectable signal has not been fully validated. In this system, once the crRNA guides the Cas12a to the target sequence, it performs cis cleavage of the target DNA and trans cleavage of the surrounding single standard reporters tagged with fluorophores or lateral flow moieties, whose readouts are essential to confirm the diagnostic potential of the sensor. These readouts could not be completed primarily due to time limitations.

Another important consideration is that, strains of the same species can have significantly different probiotic properties. However, strain level differentiation was not a primary goal of this project, it represents a valuable future direction for enhancing the diagnostic resolution.

6.6. Future Directions

A potential extension of this work involves combining a lateral flow assay, with the CRISPR-Cas12a detection technology. This method involves, extracting DNA from the sample and combining it with CRISPR RNA (crRNA), targeting the 136 bp region unique to *L. agilis* along with cas12a. Recombinant Polymerase Reagents are also added to amplify the target, along with single stranded reporter molecules. This reporter molecules get cleaved because of collateral cleavage activity of cas12a triggered by recognition and binding to target sequence.

The reporter is a two bound molecular tag, when Cas12a cleaved the reporter, the tags are separated. This reaction mixture will be applied to a lateral flow strip, where capillary action triggers migration along the membrane. Intact reporters will bind to the first capture line when no target is present, producing a single line in the strip and when the target is present, the reporter cleavage will result in two lines in the strip (**Figure 15**). This binary readout enables clear visual detection, without the need for any specialized equipment, making this assay perfect for field-based applications (62) (63) (64) .

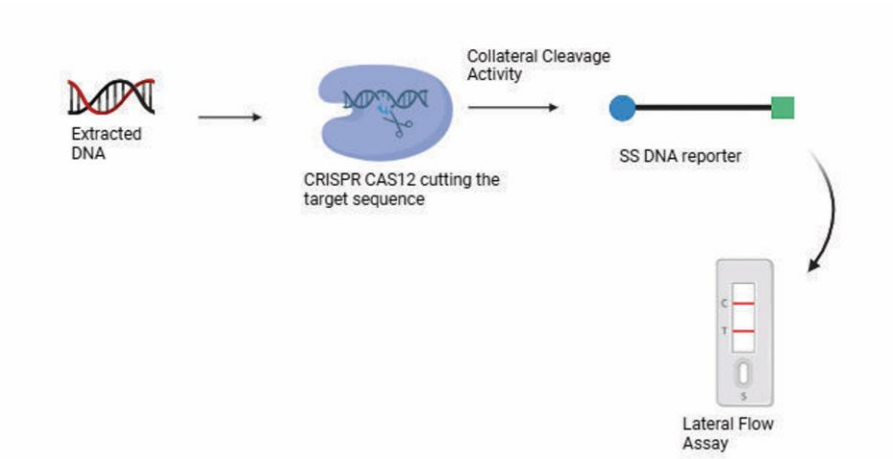
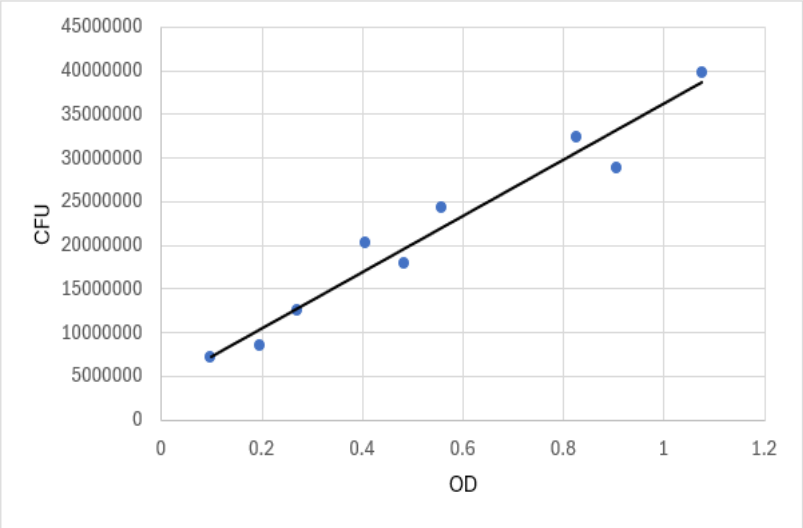


Figure 15: Collateral Cleavage Driven Lateral Flow Detection (Cas12a)

Biosensor can be further optimized by incorporation of stabilizers to improve dye stability and shelf life under fluctuating storage conditions. For strain level detection, Cas12a based RNA sensing strategies can be explored for live cell confirmation using mRNA markers specific to *L. agilis* strains. Additionally, the platform has the potential for multiplexing, by altering the crRNA sequence to detect other probiotic species (65) (66). Finally, application testing in real world matrices like yogurt or gut samples, will provide validation of the biosensor performance in practical conditions.

7. Conclusion:

This study effectively demonstrated a multi component strategy, for detecting and monitoring *L. agilis* through the integration of CRISPR-Cas12a molecular detection system with natural dye based colorimetric sensor. A framework for calculating viable bacterial cells was constructed by OD-CFU calibration curve created under controlled growth



y = 3E+07x + 4E+06

conditions(

Figure 4), allowing consistent biosensor fabrication without the need for repeated plating for CFU counting. Metabolic activity of the probiotics was indicated by observable color changes through the use of pH sensitive natural dyes (Anthocyanin) **(Figure 6, Figure 7).** At the molecular level, this study validates the published primers that targets the *L. agilis* 16S-23S ISR region **(Figure 9),** and designed extended primers in silico for improved sequence accuracy **(Figure 10).** Even though the experimental strain's WGS was not available at this time, it is still a crucial step for definitive sequence confirmation and crRNA design.

The results demonstrate proof of concept for an inexpensive, visually interpretable biosensing device that combines high specificity genotypic confirmation with phenotypic detection. Future research can concentrate on enhancing the dye stability, integrating multiplexing detection techniques and implementing CRISPR-Cas12a assays with lateral flow outputs for field applications. This platform serves as a basis for creating easily accessible strain level detection tools to promote probiotic research by connecting conventional microbiological techniques with contemporary molecular diagnostics.

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